

BINGYAO LI

📍 Winston Chung Hall 358, Riverside, CA 92507

✉ bingyao.li@ucr.edu

☎ +1 (412) 616-5592

🌐 libingyao.github.io

PROFESSIONAL APPOINTMENTS

July. 2025 – now

University of California, Riverside

Tenure-track Assistant Professor in Computer Science and Engineering

EDUCATION

Aug. 2020 – May 2025

University of Pittsburgh

Ph.D. in Computer Science

Advisor: Dr. Xulong Tang

Sep. 2017 – Jan. 2020

Tianjin University

M.S. in Computer Science and Technology

Advisor: Dr. Ce Yu, Graduated with Honor

Sep. 2013 – July 2017

Tianjin University

B.E. in Computer Science and Technology

Graduated with Honor

RESEARCH EXPERIENCE

May 2024 - Aug. 2024

NVIDIA Research, *Architecture Research Group*

Research Intern working on KV-cache management for LLM inference

Mentor: Dr. Aamer Jaleel

June 2019 - Aug. 2019

ICT of Chinese Academy of Science

Visiting Scholar working on benchmarking RISC-V architecture

Mentor: Dr. Yungang Bao

CONFERENCE PUBLICATIONS

* The authors contribute equally.

HPCA 2025

Yueqi Wang, **Bingyao Li**, Mohamed Tarek Ibn Ziad, Lieven Eeckhout, Jun Yang, Aamer Jaleel, Xulong Tang, “OASIS: Object-Aware Page Management for Multi-GPU Systems”, *The 31th IEEE International Symposium on High-Performance Computer Architecture*, 2025.

MICRO 2024

Bingyao Li, Yueqi Wang, Tianyu Wang, Lieven Eeckhout, Jun Yang, Aamer Jaleel, Xulong Tang, “STAR: Sub-Entry Sharing-Aware TLB for Multi-Instance GPU”, *In Proceedings of the 57th IEEE/ACM International Symposium on Microarchitecture*, 2024.

HPCA 2024

Yueqi Wang*, **Bingyao Li***, Aamer Jaleel, Jun Yang, Xulong Tang, “GRIT: Enhancing Multi-GPU Performance with Fine-Grained Dynamic Page Placement”, *The 30th IEEE International Symposium on High-Performance Computer Architecture*, 2024.

MICRO 2023

Bingyao Li, Yanan Guo, Yueqi Wang, Aamer Jaleel, Jun Yang, Xulong Tang, “IDYLL: Enhancing Page Translation in Multi-GPUs via Light Weight PTE Invalidations”, *In Proceedings of the 56th IEEE/ACM International Symposium on Microarchitecture*, 2023.

HPCA 2023	Bingyao Li , Jieming Yin, Anup Holey, Youtao Zhang, Jun Yang, Xulong Tang, “Trans-FW: Short Circuiting Page Table Walk in Multi-GPU Systems via Remote Forwarding”, <i>The 29th IEEE International Symposium on High-Performance Computer Architecture</i> , 2023.
DAC 2023	Bingyao Li , Yueqi Wang, Xulong Tang, “Orchestrated Scheduling and Partitioning for Improved Address Translation in GPUs”, <i>The 60th Design Automation Conference</i> , 2023.
WWW 2022	Bingyao Li* , Qi Xue*, Geng Yuan*, Sheng Li, Xiaolong Ma, Yanzhi Wang and Xulong Tang, “Optimizing Data Layout for Training Deep Neural Networks”, <i>The ACM Web Conference Workshop</i> , 2022.
MICRO 2021	Bingyao Li , Jieming Yin, Youtao Zhang, Xulong Tang, “Improving Address Translation in Multi-GPUs via Sharing and Spilling aware TLB Design”, <i>In Proceedings of the 54th IEEE/ACM International Symposium on Microarchitecture</i> , 2021.
ICA3PP 2018	Bingyao Li , Ce Yu, Xiaoteng Hu, Jian Xiao, Shanjiang Tang, Lianmeng Li, Bin Ma, “An Efficient Retrieval Method for Astronomical Catalog Time Series Data”, <i>The 18th International Conference on Algorithms and Architectures for Parallel Processing</i> , 2018
HPCC 2018	Xiaoteng Hu, Ce Yu, Bingyao Li , Shanjiang Tang, Jian Xiao, Yanyan Huang, “GAIDR: An Efficient Time Series Subsets Retrieval Method for Geo-Distributed Astronomical Data”, <i>The 20th IEEE International Conference on High Performance Computing and Communications</i> , 2018

JOURNAL

PASP 2019	Bingyao Li , Ce Yu, Chen Li, Xiaoteng Hu, Jian Xiao, Shanjiang Tang, Chenzhou Cui, and Dongwei Fan, “mcatCS: A Highly Efficient Cross-Matching Scheme for Multi-Band Astronomical Catalogs”, <i>Publication of the Astronomical Society of the Pacific</i> , 2019, 131(999).
EA 2019	Ce Yu, Bingyao Li , Jian Xiao, Chao Sun, Shanjiang Tang, Chongke Bi, Chenzhou Cui, and Dongwei Fan, “Astronomical Data Fusion: Recent Progress and Future Prospects - A Survey”, <i>Springer Experimental Astronomy</i> , 2019(6).

PATENTS

2024	Bingyao Li , Aamer Jaleel, Po-An Tsai, Anish Saxena, “Dynamic Key-Value Cache Scheduling for LLM Serving”, NVIDIA
2024	Anish Saxena, Po-An Tsai, Bingyao Li , Aamer Jaleel, “Restricted Expert Mixture Framework for MoE LLM Serving”, NVIDIA

HONORS & AWARDS

2025, 2024, 2023	Student Travel Grant, HPCA
2024, 2023, 2022	Student Travel Grant, MICRO
2023, 2022	CS50 Outstanding Research Fellowship, University of Pittsburgh
2023	PhD Forum at MICRO
2022	Student Travel Grant, ISCA
2020	SCI Research Fellowship, University of Pittsburgh
2019	National Scholarship, Ministry of Education of China
2019, 2017	Graduate Scholarship – First Prize, Tianjin University

RESEARCH TALKS

2025	Unleashing Multi-GPU Computing to the Next-Level at Binghamton University, UT Dallas, UC Riverside, ASU, William&Mary, UMass Amherst, Purdue University, and Duke University
2024	STAR: Sub-Entry Sharing-Aware TLB for Multi-Instance GPU at MICRO 2024, Austin, TX
2024, 2023	Towards Efficient and Salable Computing for Multi-GPUs at NVIDIA, USA at Tianjin University, China
2024	GRIT: Enhancing Multi-GPU Performance with Fine-Grained Dynamic Page Placement at HPCA 2024, Edinburgh, UK
2023	IDYLL: Enhancing Page Translation in Multi-GPUs via Light Weight PTE Invalidations at MICRO 2023, Toronto, ON
2023	Orchestrated Scheduling and Partitioning for Improved Address Translation in GPUs at DAC 2023, San Francisco, CA
2023	Trans-FW: Short Circuiting Page Table Walk in Multi-GPU Systems via Remote Forwarding at HPCA 2023, Montreal, QC
2022	Optimizing Data Layout for Training Deep Neural Networks at WWW 2022, Virtual
2021	Improving Address Translation in Multi-GPUs via Sharing and Spilling aware TLB Design at MICRO 2021, Virtual

ACADEMIC SERVICES

Organization Committee	Publication chair of ASPLOS 2026 Program Co-chair of GPGPU Workshop 2026
Program Committee	The International Symposium on Computer Architecture (ISCA 2026) The International Symposium on High-Performance Computer Architecture (HPCA 2026) The International Conference on Computer-Aided Design (ICCAD 2025) IEEE International Symposium on Workload Characterization (IISWC 2025)
Reviewer	The International Conference on Learning Representations (ICLR 2025) IEEE Transactions on Computers (TC) ACM Transactions on Architecture and Code Optimization (TACO) Transactions on Cloud Computing (TCC) Journal of Systems Architecture (JSA) Journal of Combinatorial Optimization (JOCO)
Artifact Evaluation Committee	IEEE/ACM International Symposium on Microarchitecture (MICRO 2024, 2022) ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS 2023)

TEACHING EXPERIENCE

Fall 2025	CS/EE 217: GPU Architecture and Parallel Programming, UC Riverside
Winter 2026	CS 162: Computer Architecture, UC Riverside
Spring 2026	CS 213: Multiprocessor Architecture and Programming, UC Riverside